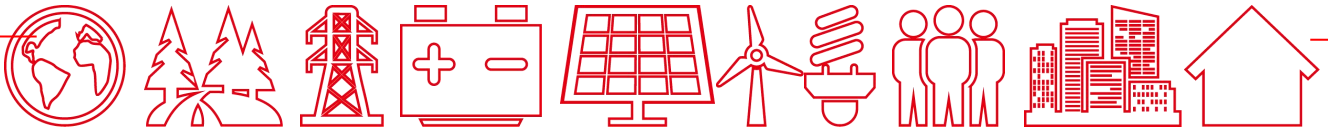


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5.7.2.3	Power quality - Verification procedure for flicker and rapid voltage changes				P
Grid impedance phase angle, φ_k [°]		30°	50°	70°	85°
Mean rated power, P/Pn [%]		Flicker, Pst			
0% Pn	L1	0.0002	0.0003	0.0002	0.0003
	L2	0.0002	0.0003	0.0003	0.0002
	L3	0.0003	0.0003	0.0003	0.0003
10% Pn	L1	0.0064	0.0062	0.0058	0.0053
	L2	0.0065	0.0062	0.0066	0.0066
	L3	0.0067	0.0065	0.0069	0.0058
20% Pn	L1	0.0067	0.0065	0.0064	0.0058
	L2	0.0070	0.0065	0.0069	0.0071
	L3	0.0068	0.0067	0.0071	0.0062
30%Pn	L1	0.0054	0.0061	0.0053	0.0057
	L2	0.0057	0.0056	0.0063	0.0057
	L3	0.0057	0.0060	0.0063	0.0057
40%Pn	L1	0.0058	0.0058	0.0056	0.0054
	L2	0.0055	0.0054	0.0051	0.0050
	L3	0.0057	0.0055	0.0054	0.0053
50%Pn	L1	0.0061	0.0064	0.0063	0.0062
	L2	0.0068	0.0056	0.0059	0.0066
	L3	0.0055	0.0061	0.0058	0.0055
60%Pn	L1	0.0050	0.0060	0.0053	0.0055
	L2	0.0057	0.0055	0.0060	0.0055
	L3	0.0048	0.0059	0.0054	0.0059
70%Pn	L1	0.0055	0.0057	0.0057	0.0054
	L2	0.0060	0.0050	0.0062	0.0059
	L3	0.0054	0.0055	0.0060	0.0058
80%Pn	L1	0.0044	0.0048	0.0048	0.0046
	L2	0.0050	0.0041	0.0049	0.0048
	L3	0.0043	0.0045	0.0047	0.0046
90%Pn	L1	0.0047	0.0059	0.0048	0.0057
	L2	0.0051	0.0049	0.0058	0.0050
	L3	0.0049	0.0055	0.0054	0.0057
100%Pn	L1	0.0064	0.0061	0.0069	0.0060

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	L2	0.0073	0.0051	0.0064	0.0071
	L3	0.0053	0.0057	0.0059	0.0062
100%Pn	L1	0.0064	0.0057	0.0068	0.0053
	L2	0.0057	0.0069	0.0063	0.0059
	L3	0.0067	0.0063	0.0064	0.0069
100%Pn	L1	0.0064	0.0056	0.0071	0.0054
	L2	0.0055	0.0067	0.0064	0.0060
	L3	0.0065	0.0063	0.0062	0.0071
Grid impedance phase angle, ϕ_k [°]		30°	50°	70°	85°
Mean rated power, P/Pn [%]		Flicker coefficient, C(ϕ_k , P)			
0% Pn	L1	0.0066	0.0099	0.0066	0.0099
	L2	0.0066	0.0099	0.0099	0.0066
	L3	0.0099	0.0099	0.0099	0.0099
10% Pn	L1	0.2112	0.2046	0.1914	0.1749
	L2	0.2145	0.2046	0.2178	0.2178
	L3	0.2211	0.2145	0.2277	0.1914
20% Pn	L1	0.2211	0.2145	0.2112	0.1914
	L2	0.2310	0.2145	0.2277	0.2343
	L3	0.2244	0.2211	0.2343	0.2046
30%Pn	L1	0.1782	0.2013	0.1749	0.1881
	L2	0.1881	0.1848	0.2079	0.1881
	L3	0.1881	0.1980	0.2079	0.1881
40%Pn	L1	0.1914	0.1914	0.1848	0.1782
	L2	0.1815	0.1782	0.1683	0.1650
	L3	0.1881	0.1815	0.1782	0.1749
50%Pn	L1	0.2013	0.2112	0.2079	0.2046
	L2	0.2244	0.1848	0.1947	0.2178
	L3	0.1815	0.2013	0.1914	0.1815
60%Pn	L1	0.1650	0.1980	0.1749	0.1815
	L2	0.1881	0.1815	0.1980	0.1815
	L3	0.1584	0.1947	0.1782	0.1947
70%Pn	L1	0.1815	0.1881	0.1881	0.1782
	L2	0.1980	0.1650	0.2046	0.1947
	L3	0.1782	0.1815	0.1980	0.1914
80%Pn	L1	0.1452	0.1584	0.1584	0.1518

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	L2	0.1650	0.1353	0.1617	0.1584
	L3	0.1419	0.1485	0.1551	0.1518
90%Pn	L1	0.1551	0.1947	0.1584	0.1881
	L2	0.1683	0.1617	0.1914	0.1650
	L3	0.1617	0.1815	0.1782	0.1881
100%Pn	L1	0.2112	0.2013	0.2277	0.1980
	L2	0.2409	0.1683	0.2112	0.2343
	L3	0.1749	0.1881	0.1947	0.2046
100%Pn	L1	0.2112	0.1881	0.2244	0.1749
	L2	0.1881	0.2277	0.2079	0.1947
	L3	0.2211	0.2079	0.2112	0.2277
100%Pn	L1	0.2112	0.1848	0.2343	0.1782
	L2	0.1815	0.2211	0.2112	0.1980
	L3	0.2145	0.2079	0.2046	0.2343

Supplementary information: The ratio of $S_{k,fc}/S_n$ used for the analysis: 33.

Rapid voltage change

Test conditions:

Case A: Switch on at any power level of primary energy

Case B: Worst case of switching of generator level

Case C: Switch on at the nominal power

Case D: Switch off at the nominal power (not emergency, but normal operational switch off)

Nominal current of PGU I_n (A)		1046			The k_{max} value:		1.049	
Test frequency (Hz)		50			--		--	
Switching action		U(V)			I _a (A)			k _i
A	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.07	689.77	689.96	549.63	556.75	551.51	0.532
	#2	690.07	689.76	689.97	545.01	552.32	546.63	0.528
	#3	690.07	689.75	689.97	544.92	551.80	546.50	0.528
A-with 0.33%	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.07	689.77	689.98	535.54	544.83	538.21	0.521
	#2	690.07	689.77	689.98	525.19	534.84	528.44	0.511
	#3	690.08	689.76	689.99	525.59	534.59	528.23	0.511
A-with 0.66%	--	L1	L2	L3	L1-L2	L2-L3	L3-L1	--
	#1	690.08	689.76	689.98	524.36	534.01	527.52	0.511
	#2	690.07	689.76	689.98	525.31	534.12	527.80	0.511
	#3	690.08	689.76	689.98	525.37	534.50	528.08	0.511
B	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.08	690.07	690.09	1044.08	1062.27	1057.89	1.016

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C	#2	689.76	689.77	689.77	1014.17	1033.11	1028.08	0.988
	#3	689.99	689.99	689.99	1075.89	1097.18	1090.31	1.049
	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.08	689.76	689.99	1072.62	1090.12	1086.02	1.042
C-with 0.33% rate	#2	690.07	689.77	689.99	1071.86	1090.40	1086.93	1.042
	#3	690.09	689.77	689.99	1071.88	1089.83	1084.34	1.042
	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.08	689.77	689.98	1041.14	1069.67	1064.29	1.023
C-with 0.66%	#2	690.07	689.77	689.99	1041.51	1069.54	1063.70	1.023
	#3	690.08	689.77	689.99	1041.35	1069.20	1063.53	1.022
	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.07	689.77	689.98	1042.26	1068.53	1063.02	1.022
D	#2	690.08	689.76	689.98	1041.75	1069.66	1063.80	1.023
	#3	690.07	689.77	689.98	1042.11	1069.94	1064.68	1.023
	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.07	689.77	689.98	1041.62	1070.58	1065.00	1.024
D	#2	690.08	689.76	689.99	1042.46	1071.58	1065.56	1.024
	#3	690.08	689.77	689.98	1042.42	1071.37	1066.12	1.024
	--	L1-L2	L2-L3	L3-L1	L1	L2	L3	--
	#1	690.07	689.77	689.98	1041.62	1070.58	1065.00	1.024

Supplementary information: Each case shall be measured three times.

5.7.2.4 Power quality - Verification procedure for DC injection							P		
Test level	33%			66%			100%		
Phase	L1	L2	L3	L1	L2	L3	L1	L2	L3
Test results (Amp)	0.1849	0.4582	0.3147	0.4304	0.5499	0.4554	0.3140	0.6912	0.6185
Test results (% of rated output current)	0.0177	0.0438	0.0301	0.0411	0.0526	0.0435	0.0300	0.0661	0.0591
Limit	≤0.5%			≤0.5%			≤0.5%		

Supplementary information: Arithmetic mean value used as test result in test duration of 5min.

- END -

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